

NETWORK PACKET ANALYSIS

(Wireshark)



HexSec Community

Introduction to Wireshark and Its Importance in Cybersecurity

Wireshark is a widely used **network protocol analyzer** that allows security professionals, system administrators, and network engineers to capture and inspect traffic in real time. It provides a **detailed view of network communications**, helping users analyze packets at a granular level.

Why Wireshark is Used in Cybersecurity?

Network Traffic Analysis

1. Helps monitor and analyze real-time data transmission across a network.
2. Identifies unusual traffic patterns that could indicate a cyber attack.

Intrusion Detection and Threat Hunting

1. Detects signs of **malicious activity**, such as unauthorized access or malware communication.
2. Assists in tracking down the **source of an attack** by analyzing traffic behavior.

Packet Inspection and Forensics

1. Examines network packets to identify **malware-infected data** or suspicious payloads.
2. Helps in post-incident investigations to understand how a breach occurred.

Performance Monitoring and Troubleshooting

1. Identifies **network congestion, dropped packets, and misconfigurations** affecting performance.
2. Troubleshoots network latency and connectivity issues efficiently.

Penetration Testing and Ethical Hacking

1. Used to analyze **vulnerabilities in network security protocols**.
2. Helps ethical hackers understand **how attackers can intercept and manipulate network traffic**.

Decrypting Encrypted Traffic (When applicable)

1. Allows inspection of **unencrypted and weakly encrypted data**.
2. Aids in ensuring sensitive information is not exposed over the network.

Wireshark is an essential tool in **cybersecurity defense**, providing visibility into network traffic to identify vulnerabilities, detect attacks, and enhance overall security.

Export Objects (Files)

Wireshark can extract files transferred through the wire. For a security analyst, it is vital to discover shared files and save them for further investigation. Exporting objects are available only for selected protocol's streams (DICOM, HTTP, IMF, SMB and TFTP).



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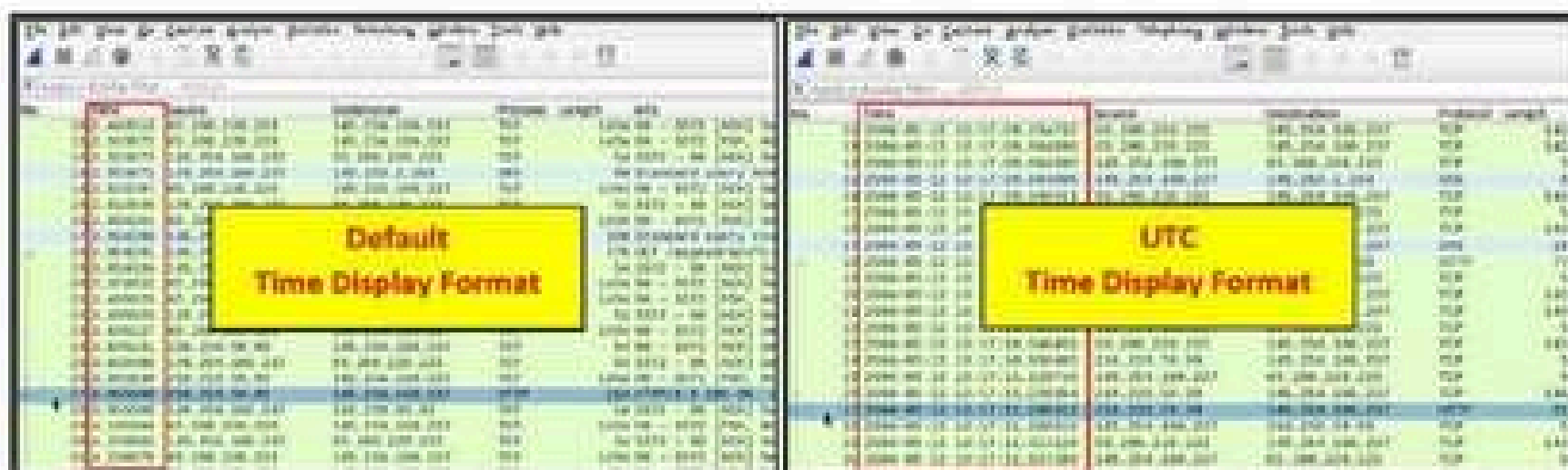
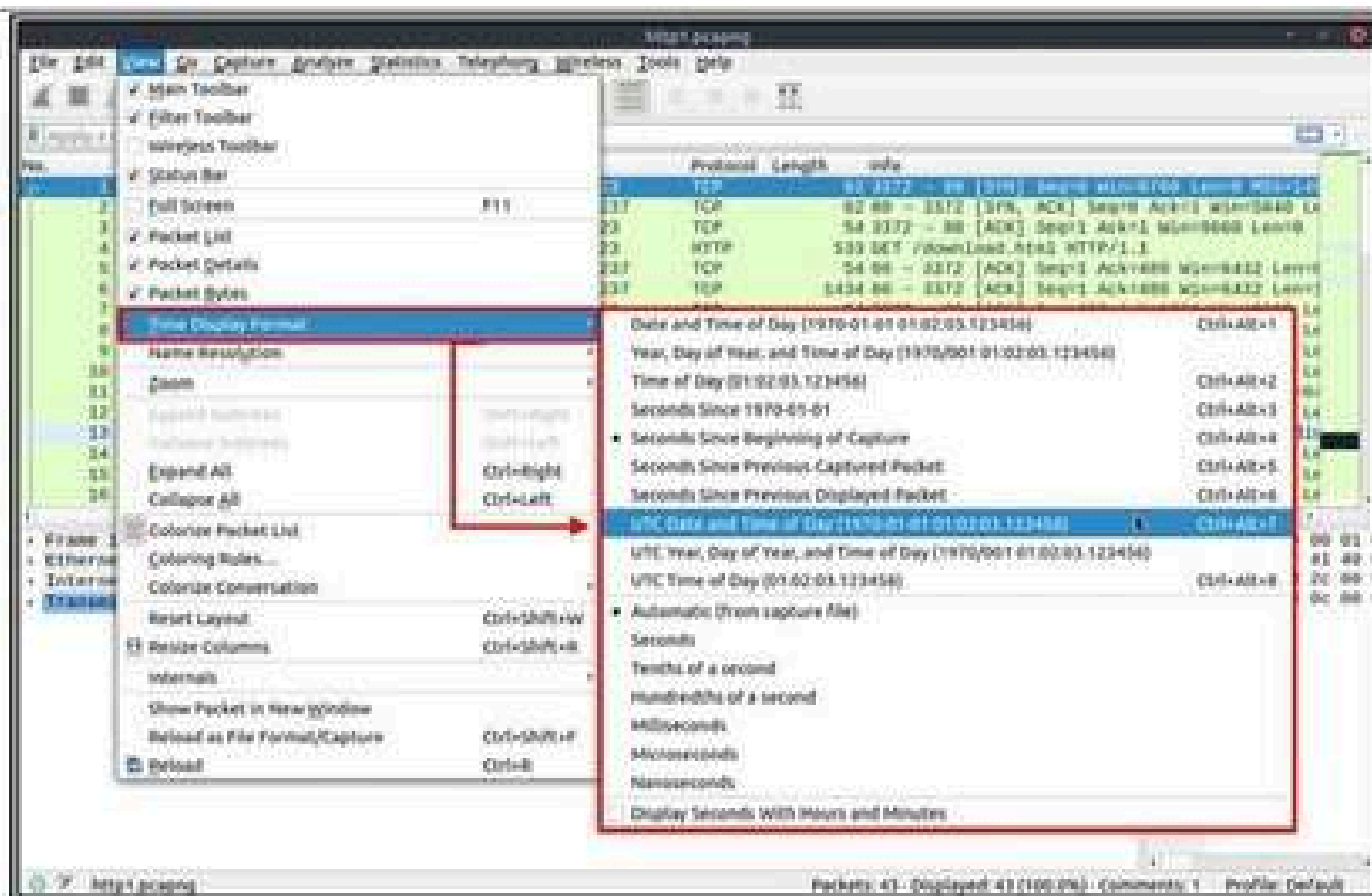


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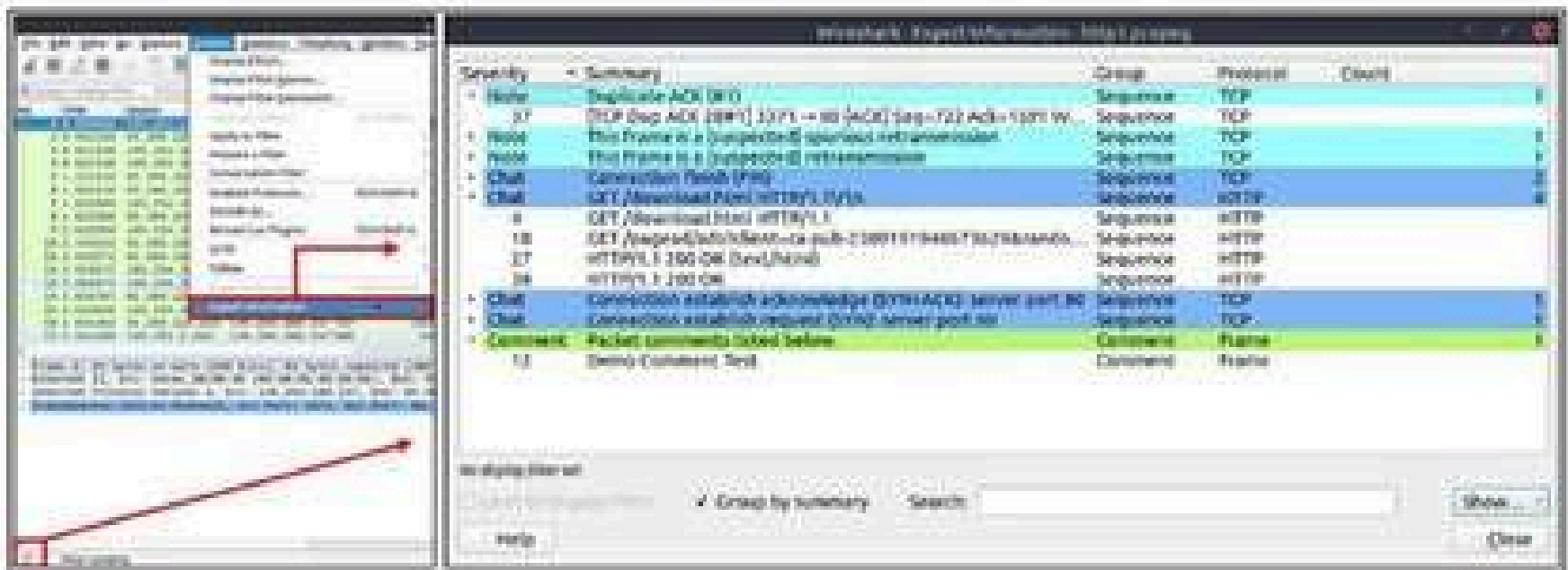
Expert Info

Wireshark also detects specific states of protocols to help analysts easily spot possible anomalies and problems. Note that these are only suggestions, and there is always a chance of having false positives/negatives. Expert info can provide a group of categories in three different severities. Details are shown in the table below.

Severity	Colour	Info
Chat	Blue	Information on usual workflow.
Note	Cyan	Notable events like application error codes.
Warn	Yellow	Warnings like unusual error codes or problem statements.
Error	Red	Problems like malformed packets.

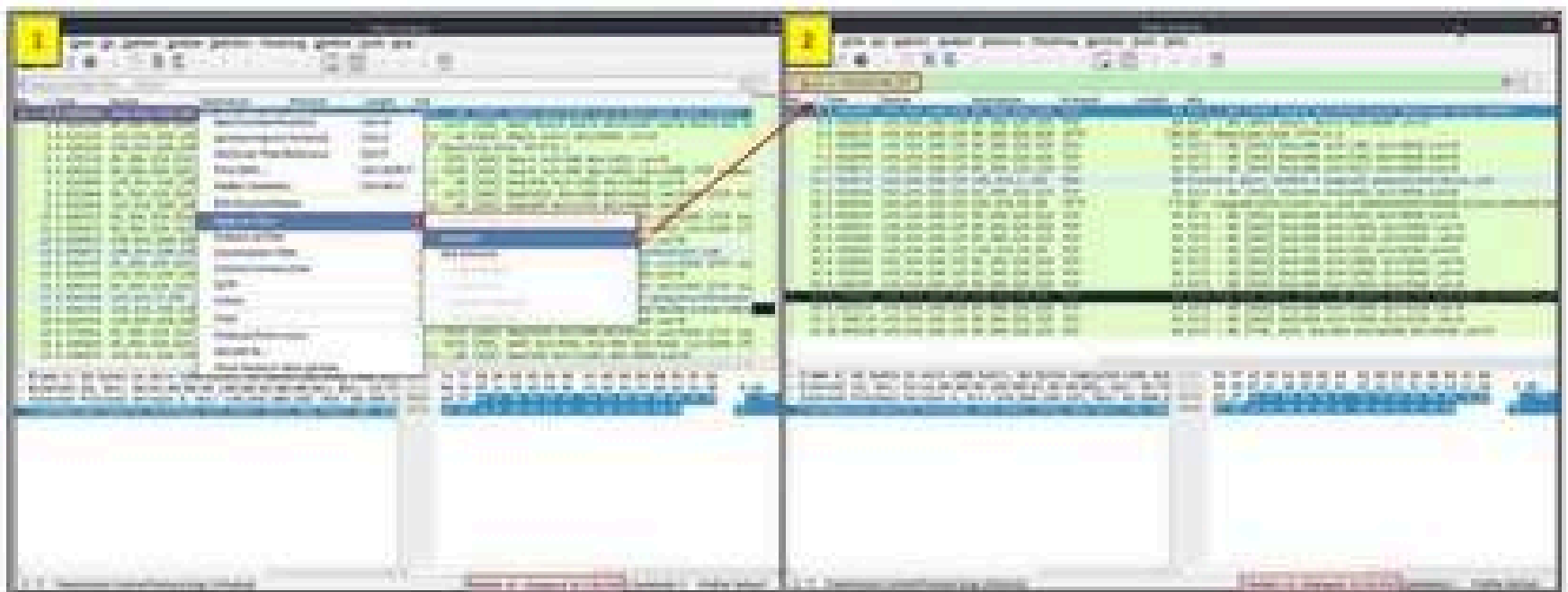
Frequently encountered information groups are listed in the table below. You can refer to Wireshark's official documentation for more information on the expert information entries.

Group	Info	Group	Info
Checksum	Checksum errors.	Deprecated	Deprecated protocol usage.
Comment	Packet comment detection.	Malformed	Malformed packet detection.



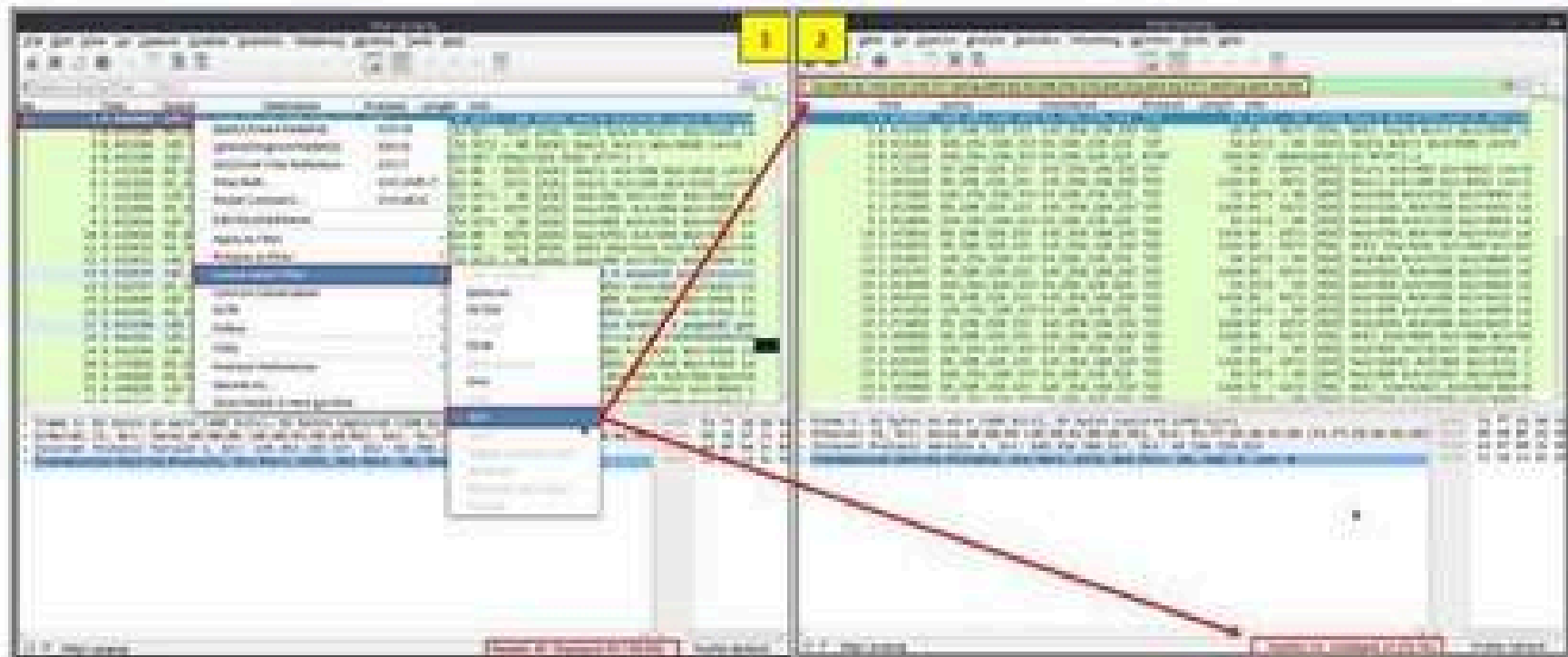
Apply as Filter

This is the most basic way of filtering traffic. While investigating a capture file, you can click on the field you want to filter and use the "right-click menu" or "Analyse --> Apply as Filter" menu to filter the specific value. Once you apply the filter, Wireshark will generate the required filter query, apply it, show the packets according to your choice, and hide the unselected packets from the packet list pane. Note that the number of total and displayed packets are always shown on the status bar.



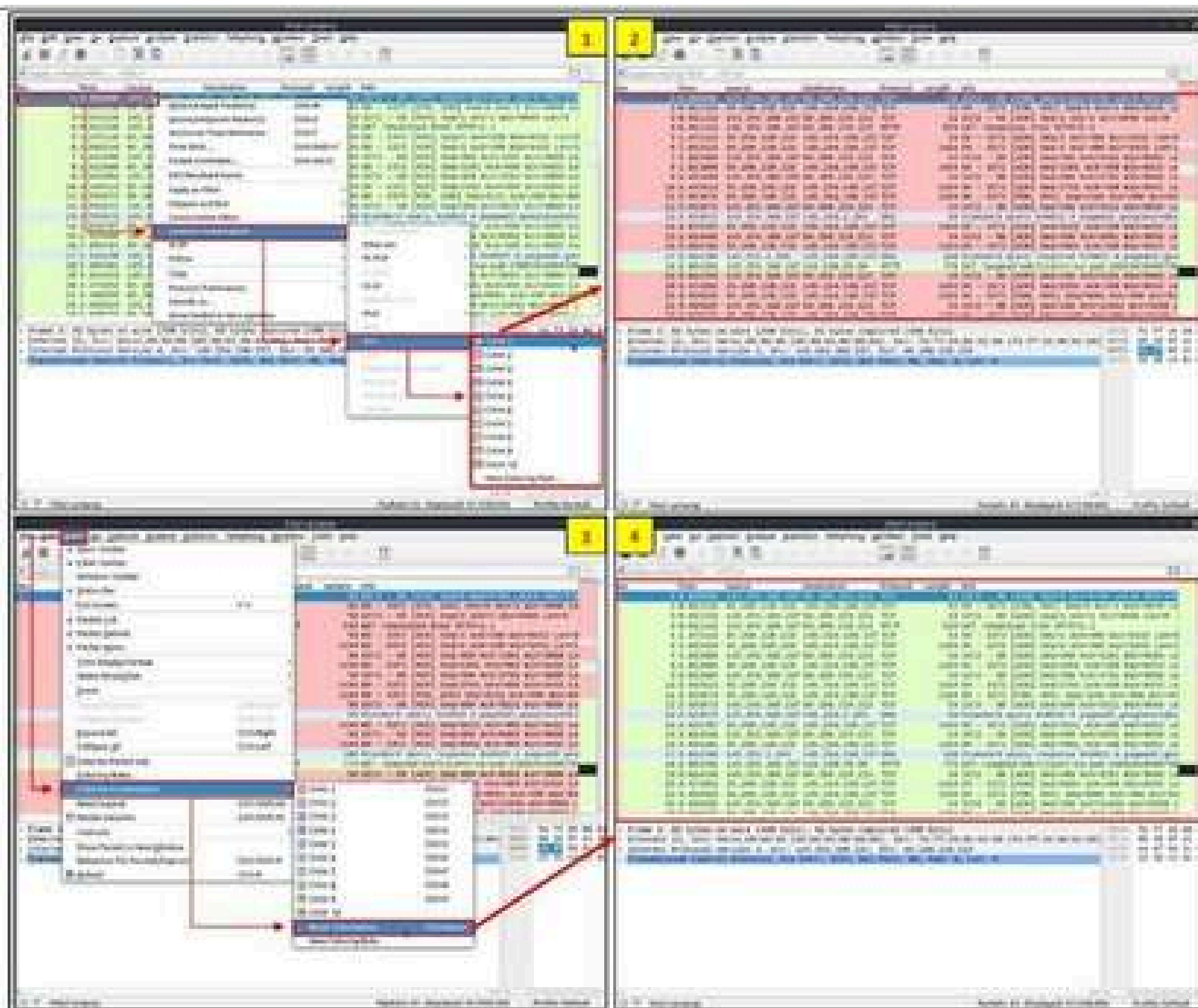
Conversation filter

- "Conversation Filter" option helps you view only the related packets and hide the rest of the packets easily. You can use the "right-click menu" or "**Analyse -> Conversation Filter**" menu to filter conversations.



Colourise Conversation

- This option is similar to the "Conversation Filter" with one difference. It highlights the linked packets without applying a display filter and decreasing the number of viewed packets. This option works with the "Colouring Rules" option and changes the packet colours without considering the previously applied colour rule. You can use the "right-click menu" or "**View --> Colourise Conversation**" menu to colourise a linked packet in a single click. Note that you can use the "View --> Colourise Conversation --> Reset Colourisation" menu to undo this operation.



Prepare as Filter

Similar to "Apply as Filter", this option helps analysts create display filters using the "right-click" menu. However, unlike the previous one, **this model doesn't apply the filters after the choice.** It adds the required query to the pane and waits for the execution command (enter) or another chosen filtering option by using the **".. and/or.."** from the "right-click menu".



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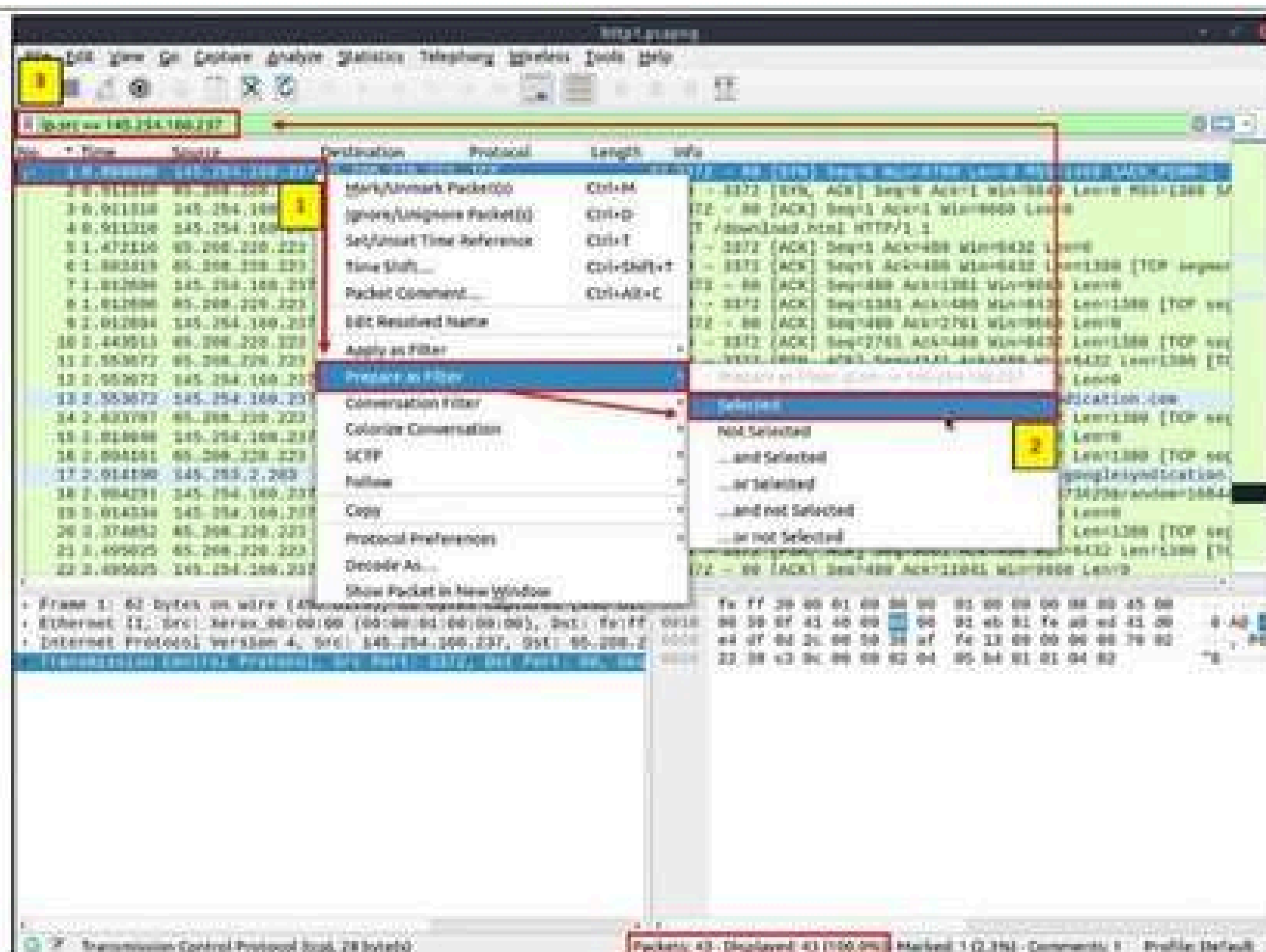


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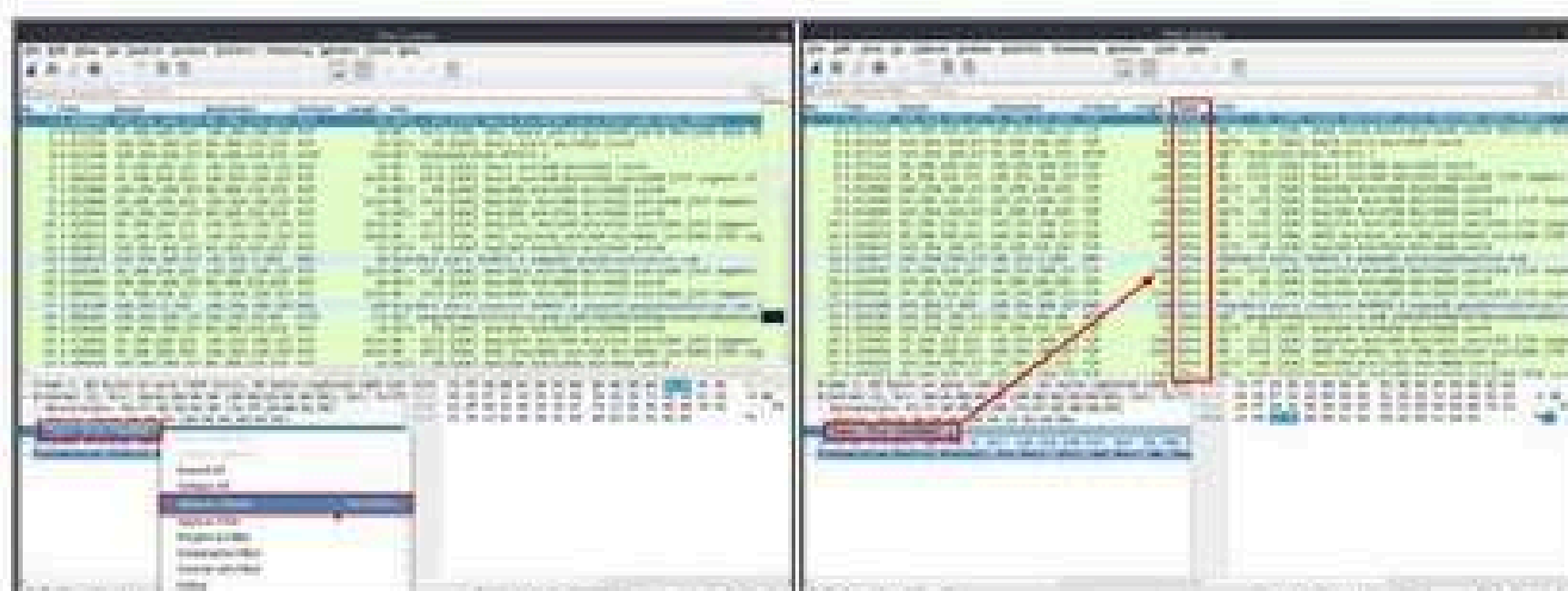
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Apply as Column

By default, the packet list pane provides basic information about each packet. You can use the "right-click menu" or "Analyse → Apply as Column" menu to add columns to the packet list pane. Once you click on a value and apply it as a column, it will be visible on the packet list pane. This function helps analysts examine the appearance of a specific value/field across the available packets in the capture file. You can enable/disable the columns shown in the packet list pane by clicking on the top of the packet list pane.



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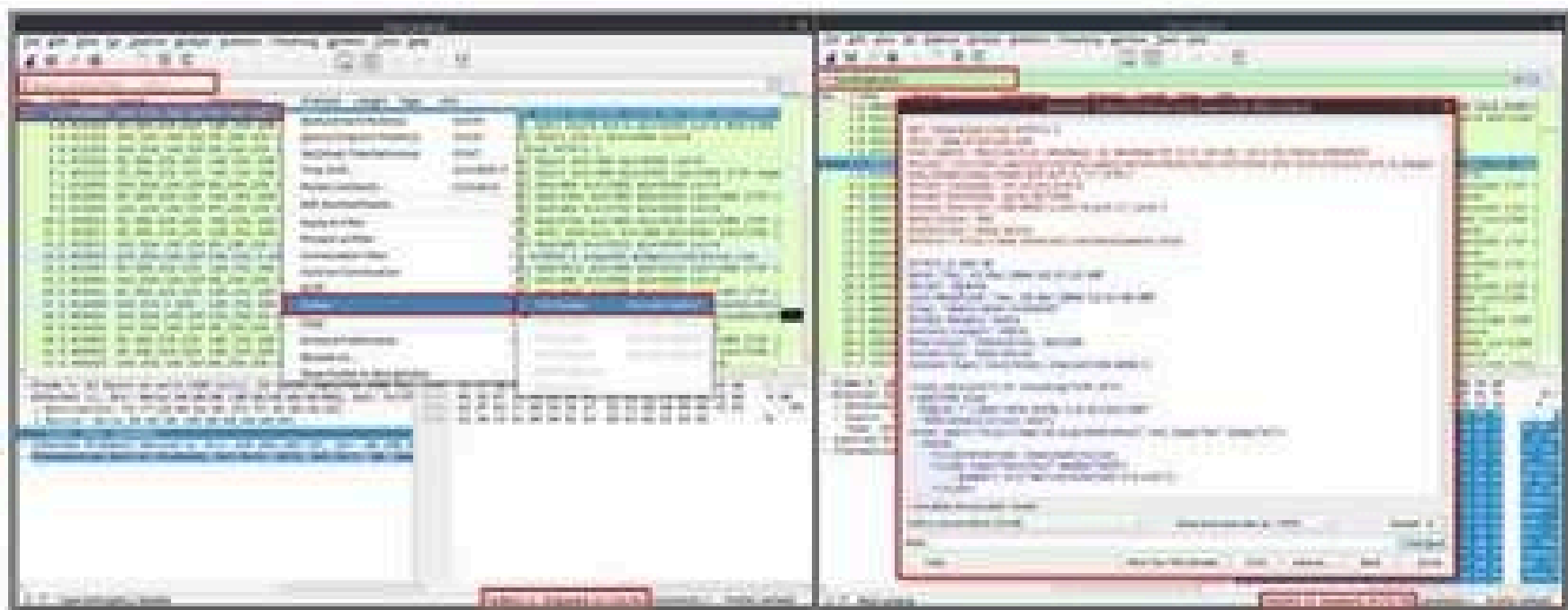


Follow Stream

Wireshark displays everything in packet portion size. However, it is possible to reconstruct the streams and view the raw traffic as it is presented at the application level. Following the protocol, streams help analysts recreate the

application-level data and understand the event of interest. It is also possible to view the unencrypted protocol data like usernames, passwords and other transferred data.

You can use the "right-click menu" or "**Analyse → Follow TCP/UDP/HTTP Stream**" menu to follow traffic streams. Streams are shown in a separate dialogue box; packets originating from the server are highlighted with blue, and those originating from the client are highlighted with red.



Once you follow a stream, Wireshark automatically creates and applies the required filter to view the specific stream. Remember, once a filter is applied, the number of the viewed packets will change. You will need to use the "**X button**" located on the right upper side of the display filter bar to remove the display filter and view all available packets in the capture file.

Statistics

- This menu provides multiple statistics options ready to investigate to help users see the big picture in terms of the scope of the traffic, available protocols, endpoints and conversations, and some protocol-specific details like DHCP, DNS and HTTP/2. For a security analyst, it is crucial to know how to utilise the static information. This section provides a quick summary of the processed pcap, which will help analysts create a hypothesis for an investigation. You can use the "**Statistics**" menu to view all available



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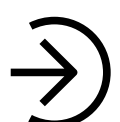
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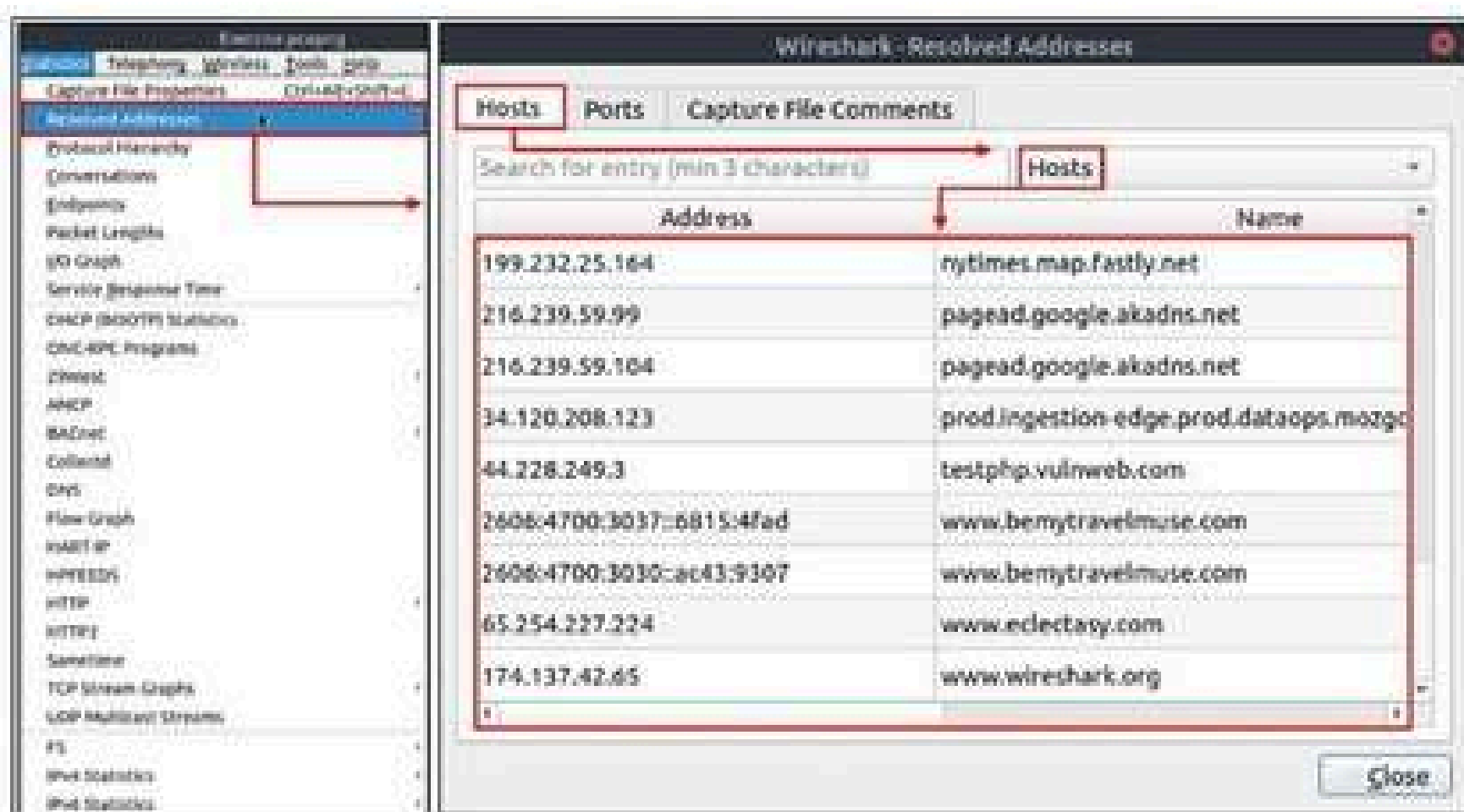


options. Now start the given VM, open the Wireshark, load the "Exercise.pcapng" file and go through the walkthrough.

Resolved Addresses

This option helps analysts identify IP addresses and DNS names available in the capture file by providing the list of the resolved addresses and their hostnames. Note that the hostname information is taken from DNS answers in the capture file. Analysts can quickly identify the accessed resources by using this menu.

Thus they can spot accessed resources and evaluate them according to the event of interest. You can use the "Statistics → Resolved Addresses" menu to view all resolved addresses by Wireshark.



Protocol Hierarchy

This option breaks down all available protocols from the capture file and helps analysts view the protocols in a tree view based on packet counters and percentages. Thus analysts can view the overall usage of the ports and services and focus on the event of interest. The golden rule mentioned in the previous

room is valid in this section; you can right-click and filter the event of interest. You can use the "Statistics → Protocol Hierarchy" menu to view this info.



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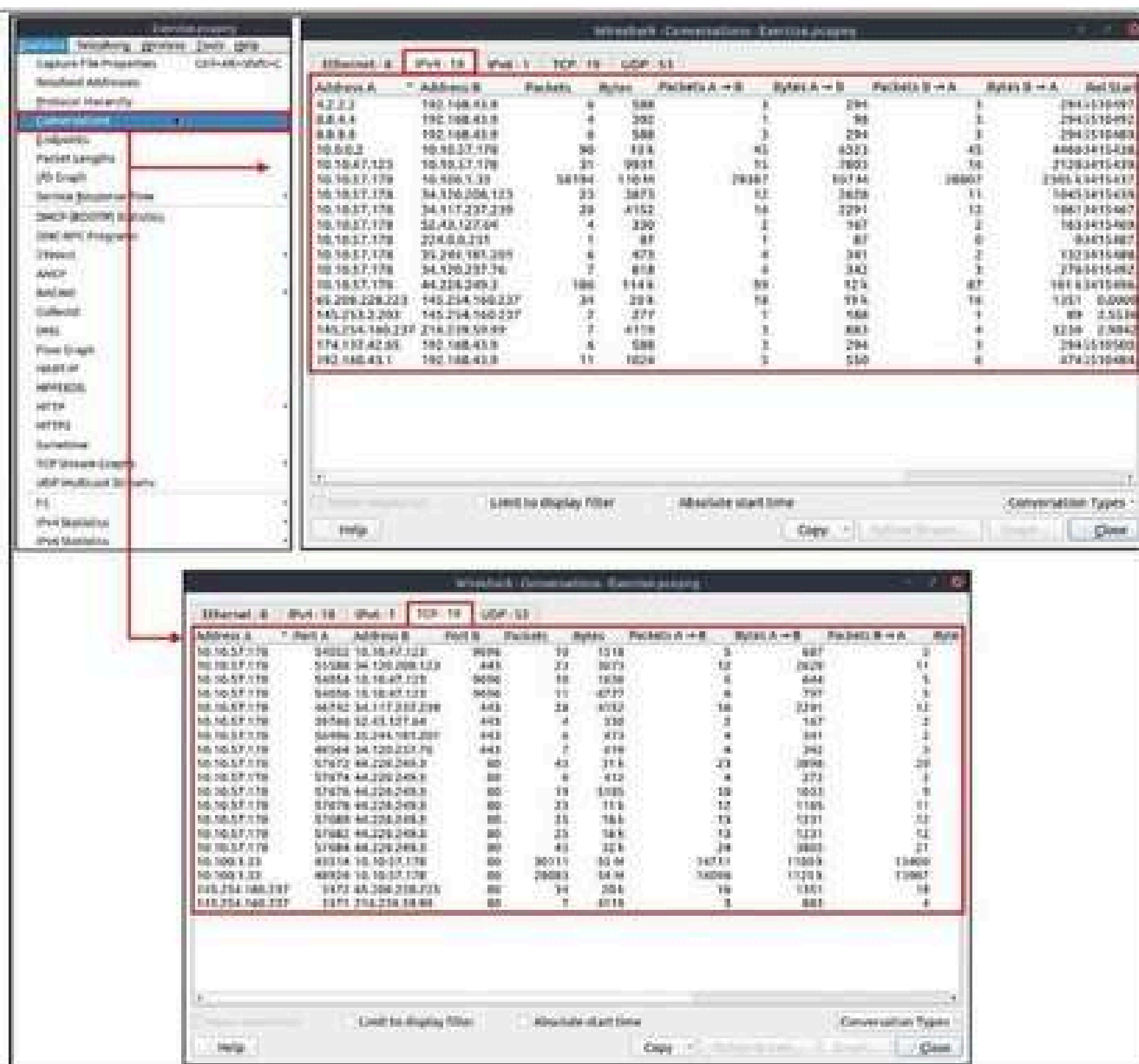


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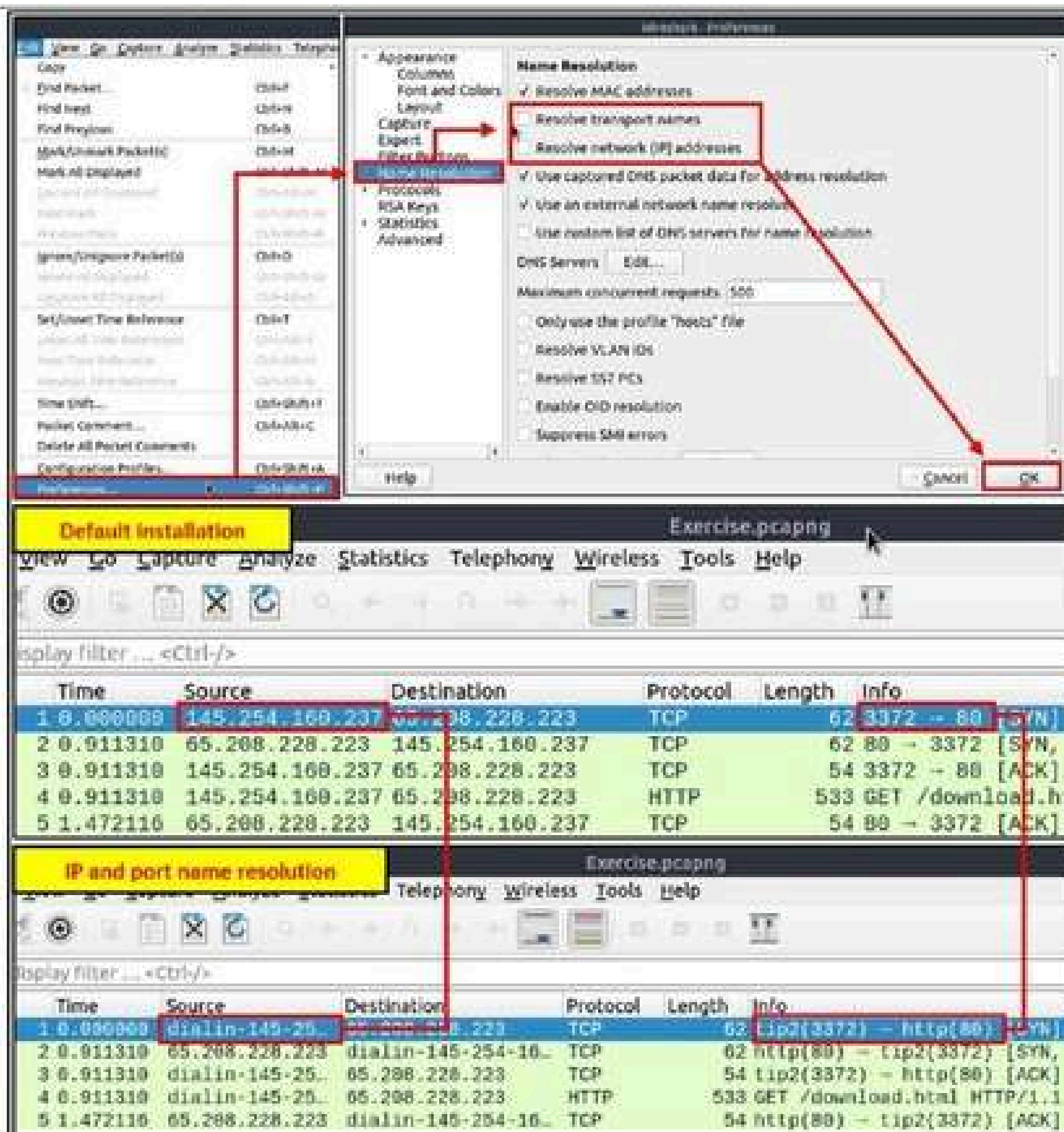


Endpoints

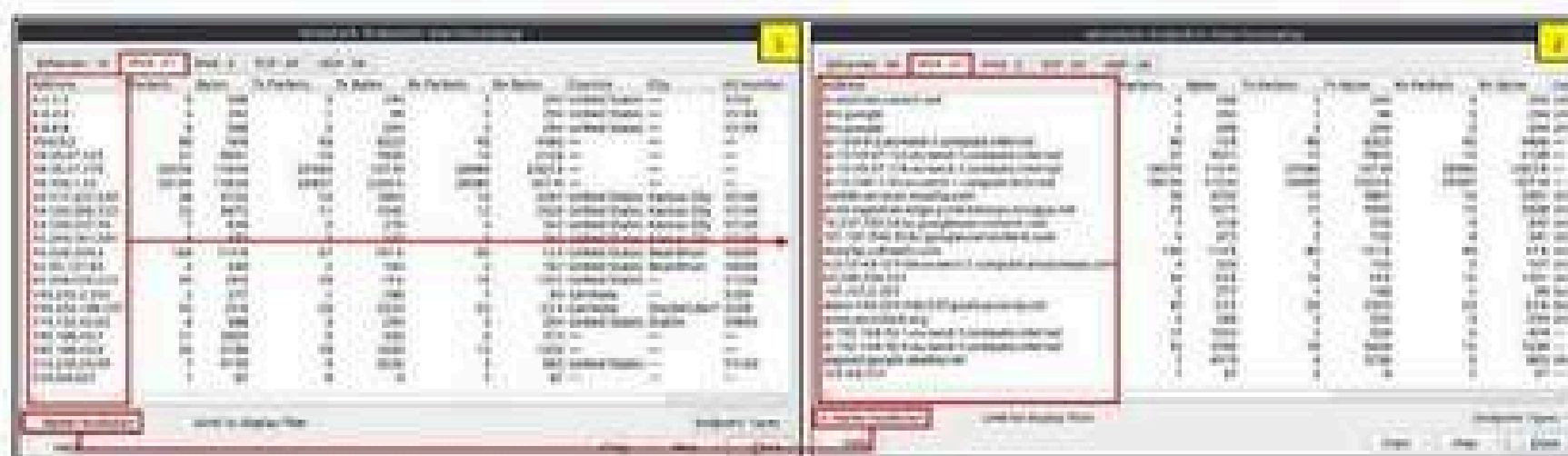
The endpoints option is similar to the conversations option. The only difference is that this option provides unique information for a single information field (Ethernet, IPv4, IPv6, TCP and UDP). Thus analysts can identify the unique endpoints in the capture file and use it for the event of interest. You can use

the "Statistics --> Endpoints" menu to view this info.

Wireshark also supports resolving MAC addresses to human-readable format using the manufacturer name assigned by IEEE. Note that this conversion is done through the first three bytes of the MAC address and only works for the known manufacturers. When you review the ethernet endpoints, you can activate this option with the "Name resolution" button in the lower-left corner of the endpoints window.



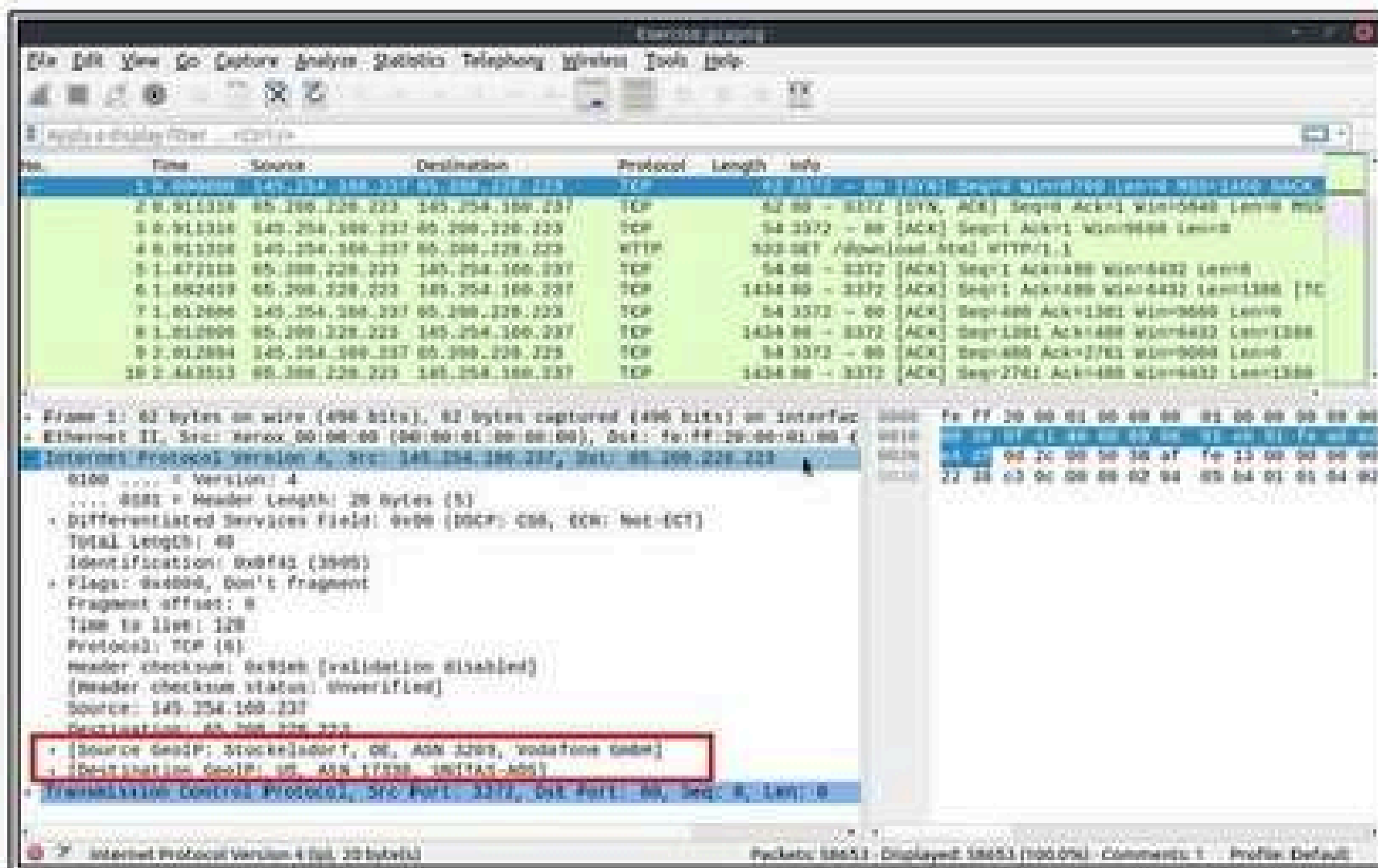
Endpoint menu view with name resolution:



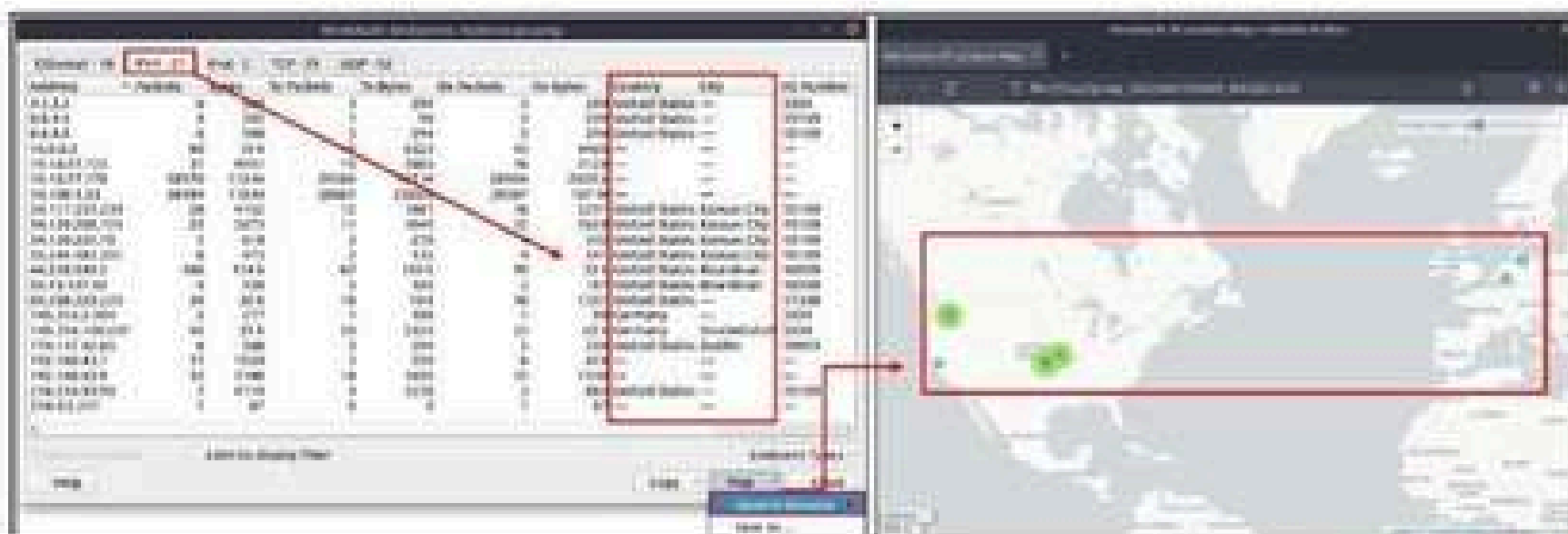
Besides name resolution, Wireshark also provides an IP geolocation mapping that helps analysts identify the map's source and destination

addresses.

But this feature is not activated by default and needs supplementary data like the **GeoIP database**. Currently, Wireshark supports MaxMind databases, and the latest versions of the Wireshark come configured MaxMind DB resolver. However, you still need MaxMind DB files and provide the database path to Wireshark by using the "Edit --> Preferences --> Name Resolution --> MaxMind database directories" menu. Once you download and indicate the path, Wireshark will automatically provide GeoIP information under the IP protocol details for the matched IP addresses.



Endpoints and GeoIP view.



IPv4 and IPv6



Up to here, almost all options provided information that contained both versions of the IP addresses. The statistics menu has two options for narrowing the statistics on packets containing a specific IP version. Thus, analysts can identify and list all events linked to specific IP versions in a single window and use it for the event of interest. You can use the "Statistics --> IPvX Statistics" menu to view this info.

The image displays four screenshots from the Wireshark network protocol analyzer, illustrating statistical analysis options:

- Top Left:** The 'Statistics' menu is open, showing 'IPvX Statistics' selected under the 'Statistics' menu.
- Top Right:** The 'IPv4 Statistics' window is shown, displaying a table of IP addresses and their statistics. The table includes columns for 'Type / Item', 'Count', 'Average', 'Min', 'Max', 'Rate (pps)', 'Percent', 'Burst rate', and 'Burst start'.
- Bottom Left:** The 'IPv6 Statistics' window is shown, displaying a table of IPv6 addresses and their statistics. The table includes columns for 'Type / Item', 'Count', 'Average', 'Min', 'Max', 'Rate (pps)', 'Percent', 'Burst rate', and 'Burst start'.
- Bottom Right:** The 'DNS Statistics' window is shown, displaying a table of DNS statistics. The table includes columns for 'Type / Item', 'Count', 'Average', 'Min', 'Max', 'Rate (pps)', 'Percent', 'Burst rate', and 'Burst start'.

DNS

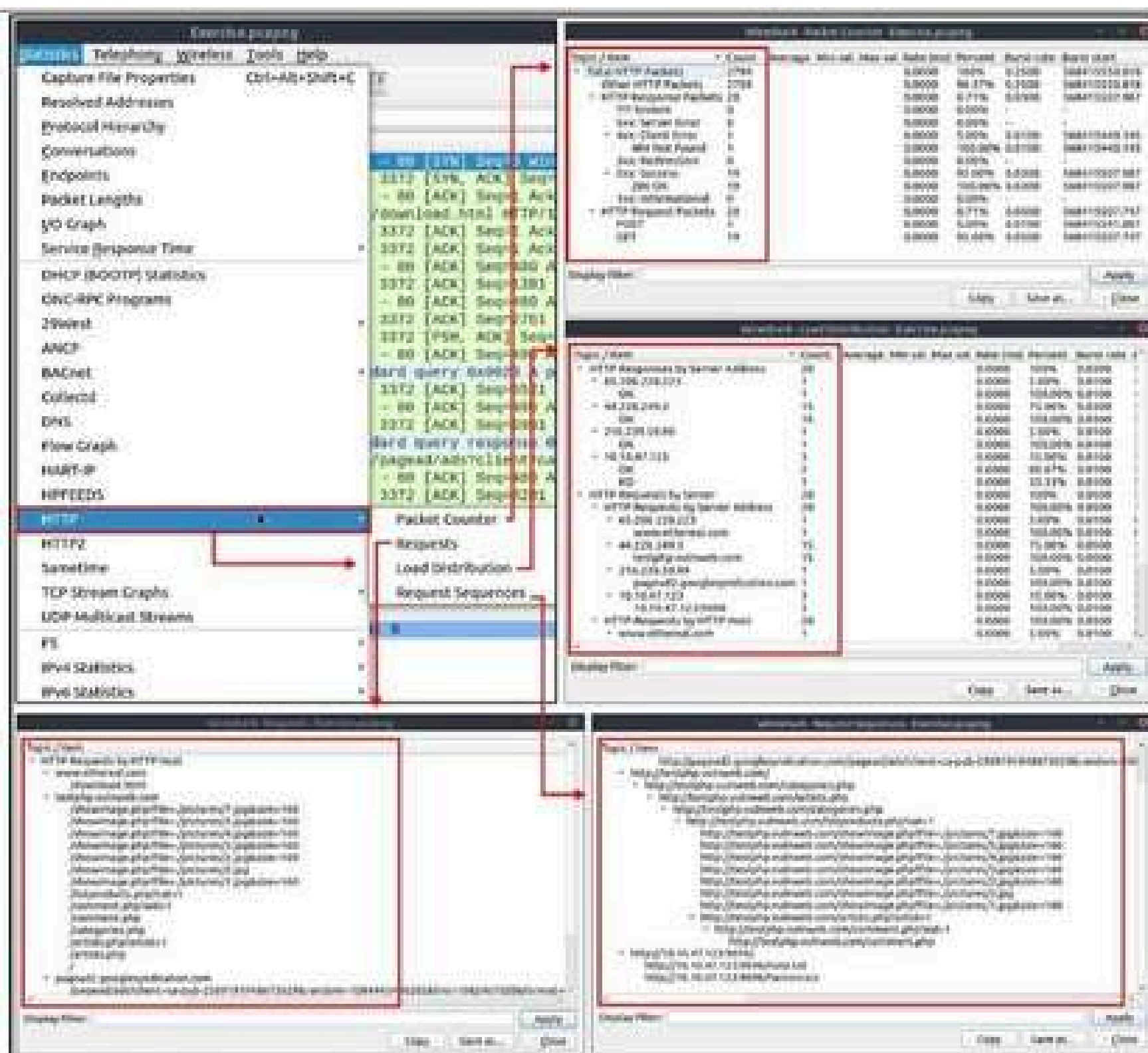
This option breaks down all DNS packets from the capture file and helps analysts view the findings in a tree view based on packet counters and percentages of the DNS protocol. Thus analysts can view the DNS service's overall usage, including rcode, opcode, class, query type, service and query stats and use it for the event of interest. You can use the "Statistics --> DNS" menu to view this info.

Exercise 20.10.1		Wireshark - DNS - Exercise 20.10.1							
Menu	Telephony	Statistics	Tools	Help					
Capture File Properties	Cut+Alt+Shift+C								
Resolved Addresses									
Protocol Hierarchy									
Conversations									
Endpoints									
Packet Lengths									
IO Graph									
Service Response Time									
DNS (SOOTPS) Statistics									
DNS RRC Programs									
DNS									
ANCH									
BACnet									
Collect									
Flow Graph									
HTTP									
HTTP2									
Sametime									
TCP Stream Graphs									
UDP Multicast Streams									
File									
IPv4 Statistics									
IPv6 Statistics									

Topic / Item	Count	Average	Min val	Max val	Rate (req)	Percent	Burst rate	Burst
Total Packets	100				0.0000	100%	0.1000	1000
No error	100				0.0000	100.00%	0.1000	1000
Standard query	100				0.0000	100.00%	0.1000	1000
Query/Response	100				0.0000	100.00%	0.1000	1000
Response	51				0.0000	49.51%	0.0800	1000
Query	52				0.0000	50.49%	0.0800	1000
Query Type	100				0.0000	100.00%	0.1000	1000
PTR (domain name pointer)	7				0.0000	6.80%	0.0200	2000
AAAA (IPv6 Address)	62				0.0000	60.19%	0.1200	1000
A (Host Address)	34				0.0000	33.01%	0.0800	1000
Class	100				0.0000	100.00%	0.1000	1000
IN	100				0.0000	100.00%	0.1000	1000
Service Stats								
request-response time (sec)	51	0.01	0.000075	0.413905	0.0000		0.0800	1000
no. of unsolicited responses	0				0.0000			
no. of retransmissions	0				0.0000			
Response Stats								
no. of questions	100	1.00	1	1	0.0000		0.1000	1000
no. of authorities	100	0.22	0	1	0.0000		0.1000	1000
no. of answers	100	0.98	0	8	0.0000		0.1000	1000
no. of additional	100	0.88	0	1	0.0000		0.1000	1000
Query Stats								
Query Len	52	27.19	9	55	0.0000		0.0800	1000
Label Stats								
4th Level or more	21				0.0000		0.0800	1000
3rd Level	20				0.0000		0.0800	1000
2nd Level	1				0.0000		0.0100	1000
1st Level	0				0.0000			
Payload size	100	75.38	35	282	0.0000	100%	0.1000	1000

HTTP

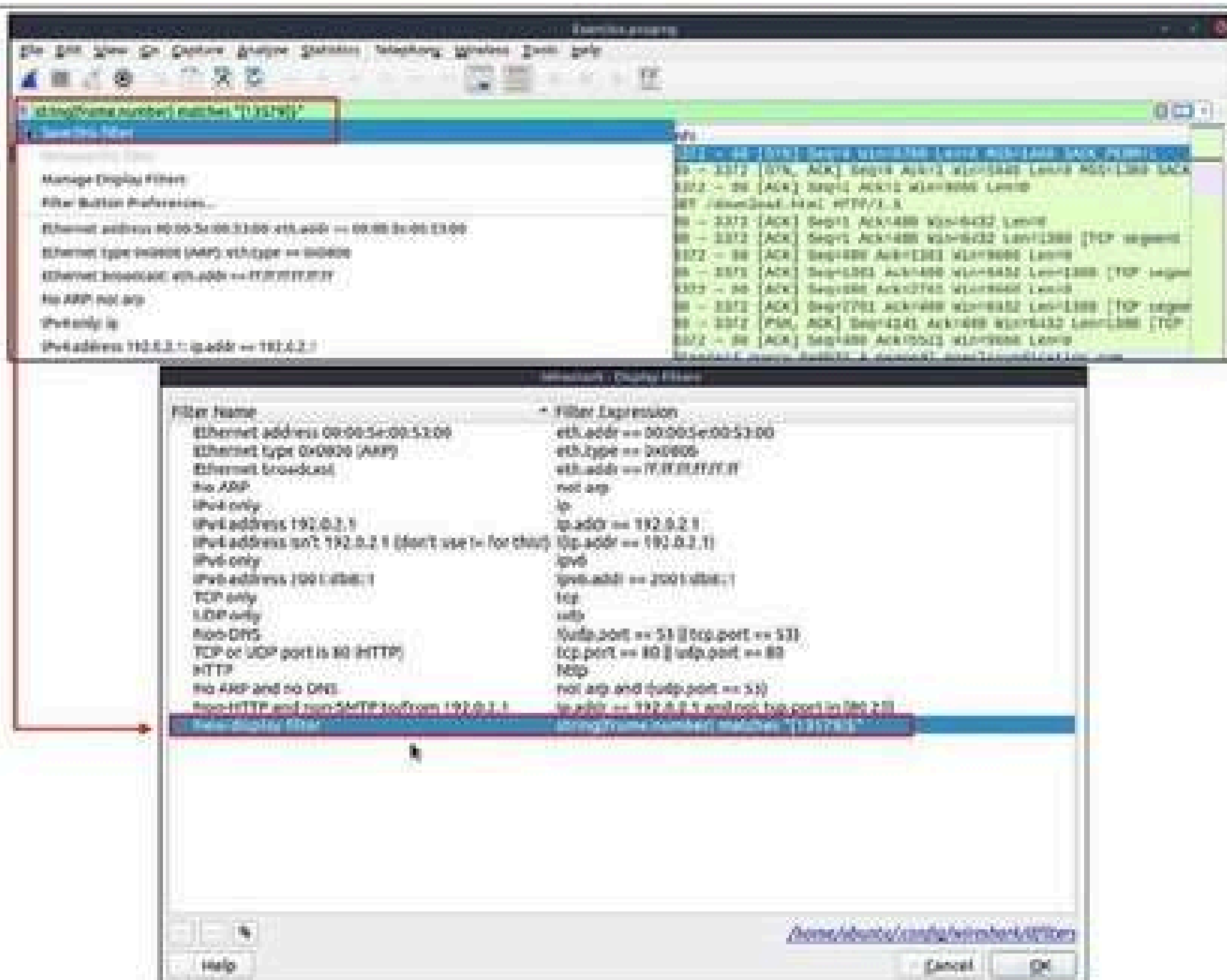
This option breaks down all HTTP packets from the capture file and helps analysts view the findings in a tree view based on packet counters and percentages of the HTTP protocol. Thus analysts can view the HTTP service's overall usage, including request and response codes and the original requests. You can use the "**Statistics --> HTTP**" menu to view this info.



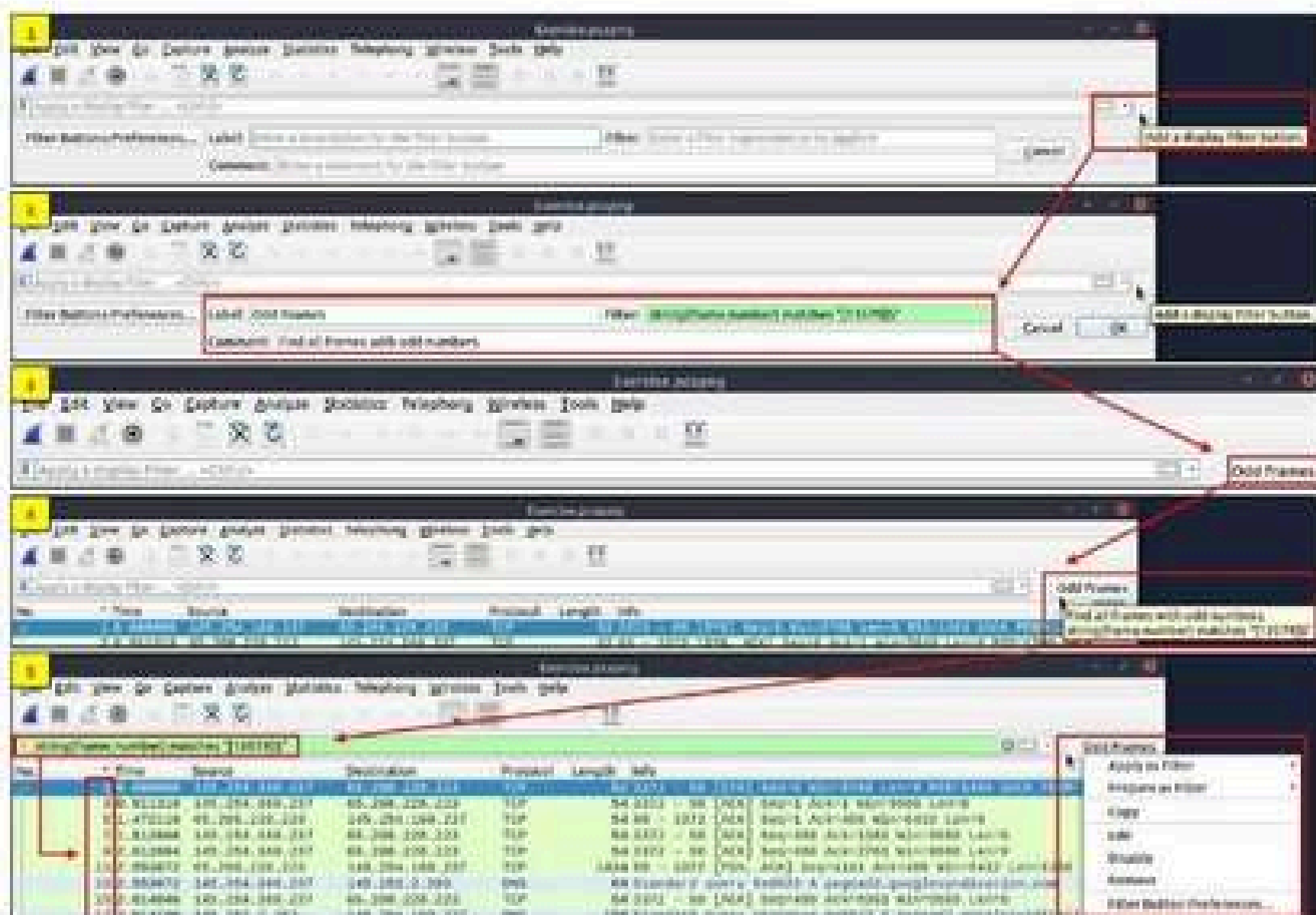
Bookmarks and Filtering Buttons

We've covered different types of filtering options, operators and functions. It is time to create filters and save them as bookmarks and buttons for later usage. As mentioned in the previous task, the filter toolbar has a filter bookmark section to save user-created filters, which helps analysts re-use favourite/complex filters with a couple of clicks. Similar to bookmarks, you can create filter buttons ready to apply with a single click.

Creating and using bookmarks.

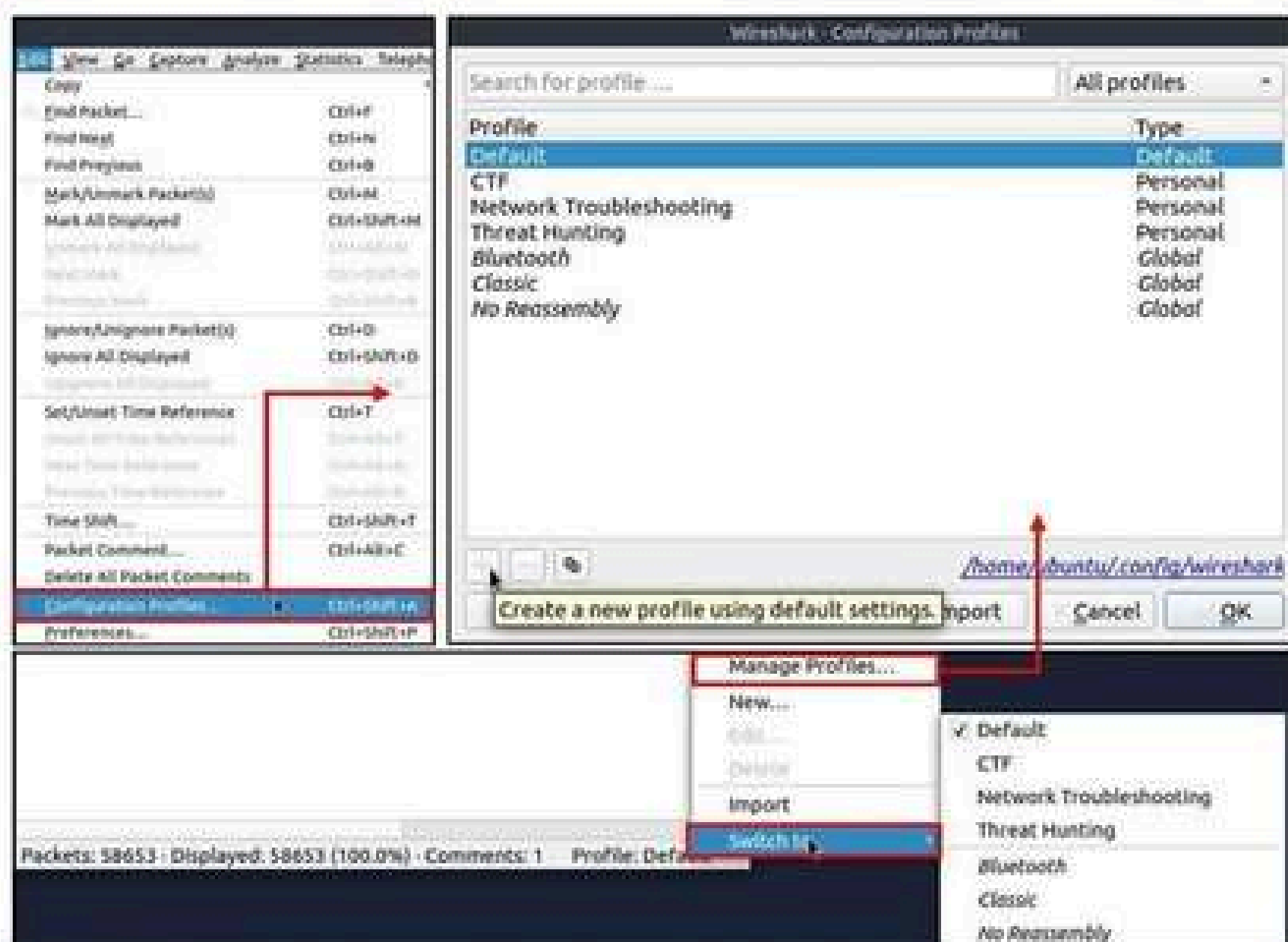


Creating and using display filter buttons.



Profiles

Wireshark is a multifunctional tool that helps analysts to accomplish in-depth packet analysis. As we covered during the room, multiple preferences need to be configured to analyse a specific event of interest. It is cumbersome to re-change the configuration for each investigation case, which requires a different set of colouring rules and filtering buttons. This is where Wireshark profiles come into play. You can create multiple profiles for different investigation cases and use them accordingly. You can use the "Edit -> Configuration Profiles" menu or the "lower right bottom of the status bar -> Profile" section to create, modify and change the profile configuration.



Packet Filtering

Capture Filters	This type of filter is used to save only a specific part of the traffic. It is set before capturing traffic and not changeable during the capture.
Display Filters	This type of filter is used to investigate packets by reducing the number of visible packets, and it is changeable during the capture.

Note: You cannot use the display filter expressions for capturing traffic and vice versa.

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